

Mohamed Ahmed

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Education

University of Minnesota

Expected May 2027

Bachelor of Science in Electrical Engineering

Relevant Coursework: Analog Electronics, Embedded Systems, Wireless Hardware Design, Antenna Design, Power Electronics, Control Systems

Experience

Michels Power

May 2026 – Aug 2026

Project Management Engineering Intern

- Supported electric utility projects through field verification, GIS records, engineering drawings, and technical documentation.
- Reviewed engineering drawings and as-built records for transmission and distribution projects.
- Developed a technical database tracking **1,200+ utility as-built records**.

Club Experience

STRIVE Medical

September 2025 – Present

Electrical / Hardware Team Lead

- Lead electrical development for a **medical robotics platform** integrating sensing, embedded control, power distribution, and actuation.
- Designed a **4-layer PCBA** supporting sensor interfaces, embedded controllers, regulated power, and motor-drive hardware.
- Integrated **Teensy 4.1, ESP32, IMUs, 14-bit encoders, and motor controllers** across robotic subsystems.
- Performed PCB bring-up and hardware debugging using oscilloscopes, logic analyzers, and multimeters.
- Led electrical development to **unanimous 1st Place** at the 2026 UMN MVP Challenge.

Projects

Wearable PPG Monitoring System | Mixed-Signal PCB, Analog Front End, Biosignal Acquisition

- Architected a mixed-signal wearable PCBA integrating a **24-bit ADC, optical analog front end, IMU, ESP32-C3, BLE**, and regulated power.
- Designed a **biased photodiode transimpedance amplifier** to convert low-level photocurrent into a measurable voltage signal.
- Designed analog signal conditioning and filtering for low-frequency PPG acquisition prior to ADC conversion.
- Integrated analog and digital subsystems on a custom PCB with attention to grounding, sensor interfaces, and low-noise signal routing.
- Performed hardware/software bring-up and debugging using oscilloscopes, logic analyzers, C, and Python.

Portable Neural Recording Device | Neural Sensing, Analog Front End Research, MedTech Hardware

- Collaborating with UMN clinicians and engineers on a portable system for recording neural signals from implanted electrodes outside the hospital.
- Investigating **low-noise amplification, signal conditioning, electrode interfaces, battery life, onboard storage, and wireless data transfer** for continuous recording.
- Evaluating hardware requirements for acquiring small neural voltage signals while minimizing noise and motion-related interference.
- Supporting early-stage architecture and technology research prior to prototype development.

Closed-Loop DC-DC Converter | Analog Sensing, Power Hardware, Circuit Modeling

- Designing a **100 W synchronous buck converter** with MOSFET power stage, gate drive, voltage/current sensing, and protection circuitry.
- Designed analog voltage and current sensing interfaces for an STM32G4-based digital controller.
- Used **LTspice and MATLAB** to evaluate circuit behavior, transient response, and control predictions prior to hardware implementation.

Skills

Analog Hardware: Analog Circuits, Transimpedance Amplifiers, Signal Conditioning, Sensor Interfaces, ADCs, Filtering, Power Regulation

PCB Design: Altium Designer, KiCad, Schematic Capture, 4-Layer PCB Design, Mixed-Signal Layout, PCB Bring-Up

Embedded: ESP32-C3, Teensy 4.1, STM32G4, Embedded C/C++, BLE, I2C, SPI, UART

Debugging & Test: Oscilloscope, Logic Analyzer, Multimeter, Root Cause Analysis, Hardware V&V

Programming: C/C++, Python, MATLAB, Verilog

Tools: Altium Designer, KiCad, LTspice, Git, ANSYS, SolidWorks